Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: Double Cross #6 (Formerly Larry B. Peterson Dairy)

Physical address of dairy:

Number and Street	City	County	Zip Code
21189 American AVE	Hilmar	Merced	95324

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 01/01/1950

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

X045-X170-X019-XXXX

B. OPERATORS

Ahlem, Mark

Operator name: Ahlem, Mark

Telephone no.: (209) 668-0867 (Landline) (Cellular)

Mailing Address Number and Street	City	State	Zip Code
23546 American AVE	Hilmar	CA	95324

This operator is responsible for paying permit fees.

C. OWNERS

Peterson, Mike or Randy

Legal owner name: Peterson, Mike or Randy

Telephone no.: (209) 667-5888 (Landline) (Cellular)

Mailing Address Number and Street	City	State	Zip Code
20949 Bloss AVE	Hilmar	CA	95324

AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	0	210	339	0	0	0
Number under roof	1,119	0	0	0	0	0
Maximum number	1,119	210	339	0	0	0
Average number	1,119	210	339	0	0	0
Avg live weight (lbs)	1,400	1,500	1,100	0		

Predominant milk cow breed: Holstein

Average milk production: 73 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 35,865.90 *tons per reporting period*

Total nitrogen from manure: 450,937.61 *lbs per reporting period*

After ammonia losses (30% loss applied): 315,656.33 *lbs per reporting period*

Total phosphorus from manure: 74,984.80 *lbs per reporting period*

Total potassium from manure: 216,504.37 *lbs per reporting period*

Total salt from manure: 575,170.65 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 29,952,000 *gallons*

Total nitrogen generated: 80,619.16 *lbs*

Total phosphorus generated: 15,033.07 *lbs*

Total potassium generated: 119,776.18 *lbs*

Total salt generated: 777,179.53 *lbs*

	27,423,000	*gallons applied*
+	2,529,000	*gallons exported*
-	0	*gallons imported*
=	29,952,000	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
T.I.D Canal	Surface water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

No dry manure nutrient imports entered.

No process wastewater nutrient imports entered.

No commercial or other nutrient imports entered.

G. NUTRIENT EXPORTS

Date	Material type	Quantity	Reporting basis	Moisture (%)	Density (lbs/cu ft)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/27/2023	Separator solids	2,000.00 *ton*	Dry-weight	55.0		5,400.00	2,200.00	4,500.00		75.10

Date	Material type	Quantity	Kjeldahl-N (mg/L)	Ammonium-N (mg/L)	Ammonia-N (mg/L)	Nitrate-N (mg/L)	P (mg/L)	K (mg/L)	EC (µmhos/cm)	TDS (mg/L)
03/02/2023	Process wastewater	255,000.00 *gal*	297.00	176.00	0.00	0.00	65.50	418.00		3,640
04/15/2023	Process wastewater	240,000.00 *gal*	549.00	381.00	0.00	0.00	43.80	1,010.00		5,860
07/22/2023	Process wastewater	510,000.00 *gal*	230.00	123.00	0.00	0.00	66.70	260.00		1,440
08/25/2023	Process wastewater	744,000.00 *gal*	230.00	123.00	0.00	0.00	66.70	260.00		1,440
09/03/2023	Process wastewater	390,000.00 *gal*	230.00	123.00	0.00	0.00	66.70	260.00		1,440
09/18/2023	Process wastewater	390,000.00 *gal*	230.00	123.00	0.00	0.00	66.70	260.00		1,440

Material type	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Dry manure	9,720.00	3,960.00	8,100.00	1,351,800.00
Process wastewater	5,635.50	1,359.25	7,325.49	43,924.41
Total exports for all materials	15,355.50	5,319.25	15,425.49	1,395,724.41

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
CSI	44	44	2	process wastewater	X045-X200-X058-XXXX
Lopes	36	36	2	process wastewater	X045-X170-X040-XXXX X045-X170-X050-XXXX X045-X170-X051-XXXX
Morgan	7	7	2	process wastewater	X045-X170-X022-XXXX
Nelson	17	17	2	process wastewater	X045-X170-X014-XXXX X045-X170-X015-XXXX
Perez	12	12	2	process wastewater	X045-X170-X035-XXXX
Rancheria	9	9	2	process wastewater	X045-X170-X018-XXXX
Randy	25	25	2	process wastewater	X045-X170-X021-XXXX
Totals for areas that were used for application	150	150	14		
Totals for areas that were not used for application					
Land application area totals	150	150	14		

B. CROPS AND HARVESTS

CSI

Field name: CSI

10/18/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 44 Plant date: 10/18/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/20/2023	884.00 *ton*	As-is		75.6	4,900.00	1,000.00	10,100.00		8.90

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	20.09	196.89	40.18	405.84	872.59

CSI

06/12/2023: Corn, silage

Crop: Corn, silage Acres planted: 44 Plant date: 06/12/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/04/2023	996.40 *ton*	As-is		58.7	4,800.00	800.00	5,100.00		5.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	22.65	217.40	36.23	230.98	935.26

Lopes

Field name: Lopes

10/17/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 36 Plant date: 10/17/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/19/2023	698.00 *ton*	As-is		65.0	5,600.00	1,300.00	11,900.00		8.70

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	19.39	217.16	50.41	461.46	1,180.78

06/09/2023: Corn, silage

Crop: Corn, silage Acres planted: 36 Plant date: 06/09/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/01/2023	959.39 *ton*	As-is		59.9	4,800.00	800.00	5,200.00		4.40

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	26.65	255.84	42.64	277.16	940.42

Morgan

Field name: Morgan

10/17/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 7 Plant date: 10/17/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/20/2023	98.00 *ton*	As-is		65.2	6,700.00	1,400.00	14,300.00		9.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	14.00	187.60	39.20	400.40	876.96

06/08/2023: Corn, silage

Crop: Corn, silage Acres planted: 7 Plant date: 06/08/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/30/2023	144.69 *ton*	As-is		55.8	6,100.00	700.00	5,300.00		4.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	20.67	252.17	28.94	219.10	767.44

Nelson

Field name: Nelson

10/17/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 17 Plant date: 10/17/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/19/2023	326.00 *ton*	As-is		65.8	4,300.00	1,100.00	9,900.00		8.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	19.18	164.92	42.19	379.69	1,049.34

Nelson

06/12/2023: Corn, silage

Crop: Corn, silage Acres planted: 17 Plant date: 06/12/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/04/2023	372.92 *ton*	As-is		57.7	4,900.00	700.00	4,600.00		4.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	21.94	214.98	30.71	201.82	853.68

Perez

Field name: Perez

10/18/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 12 Plant date: 10/18/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/20/2023	193.00 *ton*	As-is		65.2	6,700.00	1,400.00	14,300.00		9.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	16.08	215.52	45.03	459.98	1,007.46

06/08/2023: Corn, silage

Crop: Corn, silage Acres planted: 12 Plant date: 06/08/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/30/2023	262.23 *ton*	As-is		62.8	5,200.00	800.00	5,600.00		4.90

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	21.85	227.27	34.96	244.75	796.65

Rancheria

Field name: Rancheria

10/17/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 9 Plant date: 10/17/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/19/2023	145.00 *ton*	As-is		65.0	5,600.00	1,300.00	11,900.00		8.70

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	16.11	180.44	41.89	383.44	981.17

06/09/2023: Corn, silage

Crop: Corn, silage Acres planted: 9 Plant date: 06/09/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/01/2023	223.57 *ton*	As-is		62.4	4,100.00	700.00	4,500.00		4.40

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	24.84	203.70	34.78	223.57	821.94

Randy

Field name: Randy

10/18/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 25 Plant date: 10/18/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/20/2023	394.00 *ton*	As-is		65.2	6,700.00	1,400.00	14,300.00		9.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	15.76	211.18	44.13	450.74	987.21

Randy

06/08/2023: Corn, silage

Crop: Corn, silage Acres planted: 25 Plant date: 06/08/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/30/2023	536.71 *ton*	As-is		58.3	5,500.00	1,000.00	6,000.00		5.40

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	21.47	236.15	42.94	257.62	966.85

NUTRIENT BUDGET

A. LAND APPLICATIONS

CSI - 10/18/2022: Oats, silage-soft dough

Field name: CSI

Crop: Oats, silage-soft dough

Plant date: 10/18/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/10/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
T.I.D Canal	Surface water	0.00	0.00	0.00	304.21	7,460,420.00 *gal*
Application event totals		0.00	0.00	0.00	304.21	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
01/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	45.63	10.06	64.21	559.19	810,000.00 *gal*
Application event totals		45.63	10.06	64.21	559.19	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	40.56	8.94	57.08	497.06	720,000.00 *gal*
Application event totals		40.56	8.94	57.08	497.06	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	28.39	6.26	39.96	347.94	504,000.00 *gal*
Application event totals		28.39	6.26	39.96	347.94	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	57.79	4.61	106.31	616.83	555,000.00 *gal*
T.I.D Canal	Surface water	0.00	0.00	0.00	253.01	6,204,880.00 *gal*
Application event totals		57.79	4.61	106.31	869.84	

CSI - 10/18/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	39.05	3.12	71.83	381.21	375,000.00 *gal*
T.I.D Canal	Surface water	0.00	0.00	0.00	216.89	5,319,028.00 *gal*
Application event totals		39.05	3.12	71.83	598.11	

CSI - 06/12/2023: Corn, silage

Field name: CSI

Crop: Corn, silage

Plant date: 06/12/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	89.03	7.10	163.78	950.25	855,000.00 *gal*
T.I.D Canal	Surface water	0.00	0.00	0.00	266.83	6,543,800.00 *gal*
Application event totals		89.03	7.10	163.78	1,217.08	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	97.46	7.78	179.30	1,040.27	936,000.00 *gal*
T.I.D Canal	Surface water	0.00	0.00	0.00	223.29	5,476,020.00 *gal*
Application event totals		97.46	7.78	179.30	1,263.57	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
T.I.D Canal	Surface water	0.00	0.00	0.00	241.71	5,927,712.00 *gal*
Application event totals		0.00	0.00	0.00	241.71	

CSI - 06/12/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	31.41	9.11	35.50	196.64	720,000.00 *gal*
T.I.D Canal	Surface water	0.00	0.00	0.00	207.56	5,090,100.00 *gal*
Application event totals		31.41	9.11	35.50	404.20	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
T.I.D Canal	Surface water	0.00	0.00	0.00	196.42	4,817,051.00 *gal*
Application event totals		0.00	0.00	0.00	196.42	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
T.I.D Canal	Surface water	0.00	0.00	0.00	187.64	4,601,721.00 *gal*
Application event totals		0.00	0.00	0.00	187.64	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
T.I.D Canal	Surface water	0.00	0.00	0.00	158.71	3,892,116.00 *gal*
Application event totals		0.00	0.00	0.00	158.71	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	42.92	12.45	48.52	268.74	984,000.00 *gal*
T.I.D Canal	Surface water	0.00	0.00	0.00	139.64	3,424,610.00 *gal*
Application event totals		42.92	12.45	48.52	408.38	

CSI - 06/12/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	21.59	6.26	24.41	135.19	495,000.00 *gal*
T.I.D Canal	Surface water	0.00	0.00	0.00	133.02	3,262,130.00 *gal*
Application event totals		21.59	6.26	24.41	268.21	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/13/2023	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
T.I.D Canal	Surface water	0.00	0.00	0.00	122.47	3,003,421.00 *gal*
Application event totals		0.00	0.00	0.00	122.47	

Lopes - 10/17/2022: Oats, silage-soft dough

Field name: Lopes

Crop: Oats, silage-soft dough

Plant date: 10/17/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/04/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	36.58	7.25	58.00	681.51	600,000.00 *gal*
T.I.D Canal	Surface water	0.00	0.00	0.00	318.44	6,389,460.00 *gal*
Application event totals		36.58	7.25	58.00	999.95	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
12/31/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	27.43	5.43	43.50	511.13	450,000.00 *gal*
Application event totals		27.43	5.43	43.50	511.13	

Lopes - 10/17/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
01/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	39.24	8.65	55.23	480.95	570,000.00 *gal*
Application event totals		39.24	8.65	55.23	480.95	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	45.44	10.02	63.95	556.89	660,000.00 *gal*
Application event totals		45.44	10.02	63.95	556.89	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	49.63	3.96	91.31	529.77	390,000.00 *gal*
T.I.D Canal	Surface water	0.00	0.00	0.00	228.73	4,589,460.00 *gal*
Application event totals		49.63	3.96	91.31	758.50	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	24.43	1.95	44.95	260.81	192,000.00 *gal*
T.I.D Canal	Surface water	0.00	0.00	0.00	214.52	4,304,340.00 *gal*
Application event totals		24.43	1.95	44.95	475.33	

Lopes - 06/09/2023: Corn, silage

Field name: Lopes

Crop: Corn, silage

Plant date: 06/09/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

Lopes - 06/09/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	91.63	7.31	168.57	894.58	720,000.00 *gal*
T.I.D Canal	Surface water	0.00	0.00	0.00	269.64	5,410,368.00 *gal*
Application event totals		91.63	7.31	168.57	1,164.23	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
T.I.D Canal	Surface water	0.00	0.00	0.00	249.71	5,010,368.00 *gal*
Application event totals		0.00	0.00	0.00	249.71	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	26.39	7.65	29.83	165.23	495,000.00 *gal*
T.I.D Canal	Surface water	0.00	0.00	0.00	210.06	4,214,928.00 *gal*
Application event totals		26.39	7.65	29.83	375.30	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	59.18	17.16	66.90	370.52	1,110,000.00 *gal*
T.I.D Canal	Surface water	0.00	0.00	0.00	153.23	3,074,484.00 *gal*
Application event totals		59.18	17.16	66.90	523.74	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
T.I.D Canal	Surface water	0.00	0.00	0.00	190.85	3,829,460.00 *gal*
Application event totals		0.00	0.00	0.00	190.85	

Lopes - 06/09/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	32.63	9.46	36.88	204.29	612,000.00 *gal*
T.I.D Canal	Surface water	0.00	0.00	0.00	166.60	3,342,890.00 *gal*
Application event totals		32.63	9.46	36.88	370.89	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
T.I.D Canal	Surface water	0.00	0.00	0.00	160.24	3,215,180.00 *gal*
Application event totals		0.00	0.00	0.00	160.24	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	48.62	14.10	54.97	304.43	912,000.00 *gal*
T.I.D Canal	Surface water	0.00	0.00	0.00	135.64	2,721,562.00 *gal*
Application event totals		48.62	14.10	54.97	440.06	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	27.99	8.12	31.64	175.25	525,000.00 *gal*
T.I.D Canal	Surface water	0.00	0.00	0.00	141.01	2,829,431.00 *gal*
Application event totals		27.99	8.12	31.64	316.26	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
T.I.D Canal	Surface water	0.00	0.00	0.00	126.93	2,546,778.00 *gal*
Application event totals		0.00	0.00	0.00	126.93	

Morgan - 10/17/2022: Oats, silage-soft dough

Field name: Morgan

Crop: Oats, silage-soft dough

Plant date: 10/17/2022

Morgan - 10/17/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/08/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
T.I.D Canal	Surface water	0.00	0.00	0.00	292.82	1,142,435.00 *gal*
Application event totals		0.00	0.00	0.00	292.82	

12/18/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	47.03	9.32	74.57	876.23	150,000.00 *gal*
Application event totals		47.03	9.32	74.57	876.23	

02/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	42.49	9.37	59.80	520.73	120,000.00 *gal*
Application event totals		42.49	9.37	59.80	520.73	

03/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	21.24	4.69	29.90	260.36	60,000.00 *gal*
Application event totals		21.24	4.69	29.90	260.36	

04/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	49.09	3.92	90.30	523.95	75,000.00 *gal*
T.I.D Canal	Surface water	0.00	0.00	0.00	233.04	909,223.00 *gal*
Application event totals		49.09	3.92	90.30	756.99	

04/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	39.27	3.13	72.24	419.16	60,000.00 *gal*
T.I.D Canal	Surface water	0.00	0.00	0.00	224.11	874,350.00 *gal*
Application event totals		39.27	3.13	72.24	643.26	

Morgan - 06/08/2023: Corn, silage

Field name: Morgan

Crop: Corn, silage

Plant date: 06/08/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	63.81	5.09	117.40	681.13	97,500.00 *gal*
T.I.D Canal	Surface water	0.00	0.00	0.00	281.57	1,098,554.00 *gal*
Application event totals		63.81	5.09	117.40	962.70	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
T.I.D Canal	Surface water	0.00	0.00	0.00	250.50	977,326.00 *gal*
Application event totals		0.00	0.00	0.00	250.50	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
T.I.D Canal	Surface water	0.00	0.00	0.00	188.44	735,193.00 *gal*
Application event totals		0.00	0.00	0.00	188.44	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	16.45	4.77	18.60	103.00	60,000.00 *gal*
T.I.D Canal	Surface water	0.00	0.00	0.00	202.21	788,923.00 *gal*
Application event totals		16.45	4.77	18.60	305.21	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	65.81	19.08	74.39	412.00	240,000.00 *gal*
T.I.D Canal	Surface water	0.00	0.00	0.00	240.18	937,064.00 *gal*
Application event totals		65.81	19.08	74.39	652.18	

Morgan - 06/08/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	32.90	9.54	37.19	206.00	120,000.00 *gal*
T.I.D Canal	Surface water	0.00	0.00	0.00	207.87	811,006.00 *gal*
Application event totals		32.90	9.54	37.19	413.87	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	45.24	13.12	51.14	283.25	165,000.00 *gal*
T.I.D Canal	Surface water	0.00	0.00	0.00	109.94	428,921.00 *gal*
Application event totals		45.24	13.12	51.14	393.19	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	46.06	13.36	52.07	288.40	168,000.00 *gal*
T.I.D Canal	Surface water	0.00	0.00	0.00	130.46	508,977.00 *gal*
Application event totals		46.06	13.36	52.07	418.86	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	16.45	4.77	18.60	103.00	60,000.00 *gal*
T.I.D Canal	Surface water	0.00	0.00	0.00	125.39	489,217.00 *gal*
Application event totals		16.45	4.77	18.60	228.39	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
T.I.D Canal	Surface water	0.00	0.00	0.00	128.34	500,721.00 *gal*
Application event totals		0.00	0.00	0.00	128.34	

Nelson - 10/17/2022: Oats, silage-soft dough

Nelson - 10/17/2022: Oats, silage-soft dough

Field name: Nelson

Crop: Oats, silage-soft dough

Plant date: 10/17/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/03/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	38.73	7.67	61.41	721.60	300,000.00 *gal*
T.I.D Canal	Surface water	0.00	0.00	0.00	336.49	3,188,248.00 *gal*
Application event totals		38.73	7.67	61.41	1,058.08	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	30.62	6.75	43.09	375.23	210,000.00 *gal*
Application event totals		30.62	6.75	43.09	375.23	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	24.06	5.31	33.86	294.82	165,000.00 *gal*
Application event totals		24.06	5.31	33.86	294.82	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	40.42	3.23	74.37	431.49	150,000.00 *gal*
T.I.D Canal	Surface water	0.00	0.00	0.00	235.13	2,227,850.00 *gal*
Application event totals		40.42	3.23	74.37	666.61	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	68.72	5.48	126.43	733.53	255,000.00 *gal*
T.I.D Canal	Surface water	0.00	0.00	0.00	210.99	1,999,183.00 *gal*
Application event totals		68.72	5.48	126.43	944.52	

Nelson - 06/12/2023: Corn, silage

Nelson - 06/12/2023: Corn, silage

Field name: Nelson

Crop: Corn, silage

Plant date: 06/12/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	76.81	6.13	141.30	819.82	285,000.00 *gal*
T.I.D Canal	Surface water	0.00	0.00	0.00	234.25	2,219,586.00 *gal*
Application event totals		76.81	6.13	141.30	1,054.08	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
T.I.D Canal	Surface water	0.00	0.00	0.00	212.75	2,015,836.00 *gal*
Application event totals		0.00	0.00	0.00	212.75	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	67.74	19.65	76.58	424.12	600,000.00 *gal*
T.I.D Canal	Surface water	0.00	0.00	0.00	211.92	2,007,926.00 *gal*
Application event totals		67.74	19.65	76.58	636.04	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	33.87	9.82	38.29	212.06	300,000.00 *gal*
T.I.D Canal	Surface water	0.00	0.00	0.00	203.44	1,927,625.00 *gal*
Application event totals		33.87	9.82	38.29	415.50	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
T.I.D Canal	Surface water	0.00	0.00	0.00	182.11	1,725,520.00 *gal*
Application event totals		0.00	0.00	0.00	182.11	

Nelson - 06/12/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	37.94	11.00	42.88	237.51	336,000.00 *gal*
T.I.D Canal	Surface water	0.00	0.00	0.00	169.39	1,605,001.00 *gal*
Application event totals		37.94	11.00	42.88	406.90	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
T.I.D Canal	Surface water	0.00	0.00	0.00	195.13	1,848,835.00 *gal*
Application event totals		0.00	0.00	0.00	195.13	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	35.23	10.22	39.82	220.54	312,000.00 *gal*
T.I.D Canal	Surface water	0.00	0.00	0.00	171.62	1,626,117.00 *gal*
Application event totals		35.23	10.22	39.82	392.16	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	33.87	9.82	38.29	212.06	300,000.00 *gal*
T.I.D Canal	Surface water	0.00	0.00	0.00	158.50	1,501,778.00 *gal*
Application event totals		33.87	9.82	38.29	370.56	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
T.I.D Canal	Surface water	0.00	0.00	0.00	150.38	1,424,899.00 *gal*
Application event totals		0.00	0.00	0.00	150.38	

Perez - 10/18/2022: Oats, silage-soft dough

Field name: Perez

Crop: Oats, silage-soft dough

Plant date: 10/18/2022

Perez - 10/18/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/10/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
T.I.D Canal	Surface water	0.00	0.00	0.00	284.47	1,902,624.00 *gal*
Application event totals		0.00	0.00	0.00	284.47	

02/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	30.98	6.83	43.60	379.70	150,000.00 *gal*
Application event totals		30.98	6.83	43.60	379.70	

02/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	25.26	3.40	23.15	131.43	52,500.00 *gal*
Application event totals		25.26	3.40	23.15	131.43	

03/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	24.78	5.47	34.88	303.76	120,000.00 *gal*
Application event totals		24.78	5.47	34.88	303.76	

04/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	73.30	5.85	134.86	782.43	192,000.00 *gal*
T.I.D Canal	Surface water	0.00	0.00	0.00	240.41	1,607,940.00 *gal*
Application event totals		73.30	5.85	134.86	1,022.84	

04/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	68.72	5.48	126.43	733.53	180,000.00 *gal*
T.I.D Canal	Surface water	0.00	0.00	0.00	217.67	1,455,876.00 *gal*
Application event totals		68.72	5.48	126.43	951.20	

Perez - 06/08/2023: Corn, silage

Field name: Perez

Crop: Corn, silage

Plant date: 06/08/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	80.17	6.40	147.50	855.78	210,000.00 *gal*
T.I.D Canal	Surface water	0.00	0.00	0.00	307.32	2,055,456.00 *gal*
Application event totals		80.17	6.40	147.50	1,163.10	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
T.I.D Canal	Surface water	0.00	0.00	0.00	284.51	1,902,891.00 *gal*
Application event totals		0.00	0.00	0.00	284.51	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	40.79	11.83	46.11	255.36	255,000.00 *gal*
T.I.D Canal	Surface water	0.00	0.00	0.00	236.81	1,583,852.00 *gal*
Application event totals		40.79	11.83	46.11	492.17	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	33.59	9.74	37.97	210.29	210,000.00 *gal*
T.I.D Canal	Surface water	0.00	0.00	0.00	202.49	1,354,288.00 *gal*
Application event totals		33.59	9.74	37.97	412.78	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
T.I.D Canal	Surface water	0.00	0.00	0.00	184.55	1,234,300.00 *gal*
Application event totals		0.00	0.00	0.00	184.55	

Perez - 06/08/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	38.39	11.13	43.39	240.34	240,000.00 *gal*
T.I.D Canal	Surface water	0.00	0.00	0.00	162.93	1,089,702.00 *gal*
Application event totals		38.39	11.13	43.39	403.26	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	40.79	11.83	46.11	255.36	255,000.00 *gal*
T.I.D Canal	Surface water	0.00	0.00	0.00	150.59	1,007,210.00 *gal*
Application event totals		40.79	11.83	46.11	405.95	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	28.79	8.35	32.55	180.25	180,000.00 *gal*
T.I.D Canal	Surface water	0.00	0.00	0.00	136.02	909,726.00 *gal*
Application event totals		28.79	8.35	32.55	316.27	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	19.19	5.57	21.70	120.17	120,000.00 *gal*
T.I.D Canal	Surface water	0.00	0.00	0.00	116.46	778,923.00 *gal*
Application event totals		19.19	5.57	21.70	236.63	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
T.I.D Canal	Surface water	0.00	0.00	0.00	90.72	606,741.00 *gal*
Application event totals		0.00	0.00	0.00	90.72	

Rancheria - 10/17/2022: Oats, silage-soft dough

Rancheria - 10/17/2022: Oats, silage-soft dough

Field name: Rancheria

Crop: Oats, silage-soft dough

Plant date: 10/17/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/05/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
T.I.D Canal	Surface water	0.00	0.00	0.00	286.89	1,439,091.00 *gal*
Application event totals		0.00	0.00	0.00	286.89	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
01/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	77.00	10.37	70.54	400.56	120,000.00 *gal*
Application event totals		77.00	10.37	70.54	400.56	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	37.18	8.20	52.32	455.64	135,000.00 *gal*
Application event totals		37.18	8.20	52.32	455.64	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	20.65	4.55	29.07	253.13	75,000.00 *gal*
Application event totals		20.65	4.55	29.07	253.13	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	53.45	4.26	98.33	570.52	105,000.00 *gal*
T.I.D Canal	Surface water	0.00	0.00	0.00	237.77	1,192,725.00 *gal*
Application event totals		53.45	4.26	98.33	808.29	

Rancheria - 10/17/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	30.54	2.44	56.19	326.01	60,000.00 *gal*
T.I.D Canal	Surface water	0.00	0.00	0.00	217.25	1,089,765.00 *gal*
Application event totals		30.54	2.44	56.19	543.26	

Rancheria - 06/09/2023: Corn, silage

Field name: Rancheria

Crop: Corn, silage

Plant date: 06/09/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	80.17	6.40	147.50	855.78	157,500.00 *gal*
T.I.D Canal	Surface water	0.00	0.00	0.00	271.34	1,361,107.00 *gal*
Application event totals		80.17	6.40	147.50	1,127.12	

06/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
T.I.D Canal	Surface water	0.00	0.00	0.00	241.75	1,212,665.00 *gal*
Application event totals		0.00	0.00	0.00	241.75	

07/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
T.I.D Canal	Surface water	0.00	0.00	0.00	201.60	1,011,265.00 *gal*
Application event totals		0.00	0.00	0.00	201.60	

Rancheria - 06/09/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	31.99	9.28	36.16	200.28	150,000.00 *gal*
T.I.D Canal	Surface water	0.00	0.00	0.00	208.70	1,046,907.00 *gal*
Application event totals		31.99	9.28	36.16	408.98	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	31.99	9.28	36.16	200.28	150,000.00 *gal*
T.I.D Canal	Surface water	0.00	0.00	0.00	200.92	1,007,873.00 *gal*
Application event totals		31.99	9.28	36.16	401.20	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	51.18	14.84	57.86	320.45	240,000.00 *gal*
T.I.D Canal	Surface water	0.00	0.00	0.00	197.30	989,721.00 *gal*
Application event totals		51.18	14.84	57.86	517.75	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	28.79	8.35	32.55	180.25	135,000.00 *gal*
T.I.D Canal	Surface water	0.00	0.00	0.00	157.73	791,199.00 *gal*
Application event totals		28.79	8.35	32.55	337.98	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	30.71	8.91	34.72	192.27	144,000.00 *gal*
T.I.D Canal	Surface water	0.00	0.00	0.00	137.20	688,211.00 *gal*
Application event totals		30.71	8.91	34.72	329.47	

Rancheria - 06/09/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	20.79	6.03	23.51	130.18	97,500.00 *gal*
T.I.D Canal	Surface water	0.00	0.00	0.00	184.63	926,153.00 *gal*
Application event totals		20.79	6.03	23.51	314.81	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
T.I.D Canal	Surface water	0.00	0.00	0.00	121.45	609,217.00 *gal*
Application event totals		0.00	0.00	0.00	121.45	

Randy - 10/18/2022: Oats, silage-soft dough

Field name: Randy

Crop: Oats, silage-soft dough

Plant date: 10/18/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/08/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	55.31	10.96	87.69	1,030.44	630,000.00 *gal*
T.I.D Canal	Surface water	0.00	0.00	0.00	297.87	4,150,550.00 *gal*
Application event totals		55.31	10.96	87.69	1,328.31	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
01/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	35.69	7.87	50.23	437.41	360,000.00 *gal*
Application event totals		35.69	7.87	50.23	437.41	

Randy - 10/18/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	17.84	3.94	25.12	218.71	180,000.00 *gal*
Application event totals		17.84	3.94	25.12	218.71	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	48.38	3.86	89.00	516.40	264,000.00 *gal*
Application event totals		48.38	3.86	89.00	516.40	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	57.73	4.61	106.20	616.16	315,000.00 *gal*
T.I.D Canal	Surface water	0.00	0.00	0.00	235.40	3,280,050.00 *gal*
Application event totals		57.73	4.61	106.20	851.56	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
T.I.D Canal	Surface water	0.00	0.00	0.00	215.52	3,003,000.00 *gal*
Application event totals		0.00	0.00	0.00	215.52	

Randy - 06/08/2023: Corn, silage

Field name: Randy

Crop: Corn, silage

Plant date: 06/08/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

Randy - 06/08/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	87.96	7.02	161.83	938.91	480,000.00 *gal*
T.I.D Canal	Surface water	0.00	0.00	0.00	307.78	4,288,550.00 *gal*
Application event totals		87.96	7.02	161.83	1,246.69	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
T.I.D Canal	Surface water	0.00	0.00	0.00	266.48	3,713,125.00 *gal*
Application event totals		0.00	0.00	0.00	266.48	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
T.I.D Canal	Surface water	0.00	0.00	0.00	231.44	3,224,900.00 *gal*
Application event totals		0.00	0.00	0.00	231.44	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	33.40	9.69	37.75	209.09	435,000.00 *gal*
T.I.D Canal	Surface water	0.00	0.00	0.00	220.43	3,071,475.00 *gal*
Application event totals		33.40	9.69	37.75	429.52	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	40.31	11.69	45.56	252.35	525,000.00 *gal*
T.I.D Canal	Surface water	0.00	0.00	0.00	205.48	2,863,125.00 *gal*
Application event totals		40.31	11.69	45.56	457.83	

Randy - 06/08/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	46.06	13.36	52.07	288.40	600,000.00 *gal*
T.I.D Canal	Surface water	0.00	0.00	0.00	141.56	1,972,440.00 *gal*
Application event totals		46.06	13.36	52.07	429.96	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	34.55	10.02	39.05	216.30	450,000.00 *gal*
T.I.D Canal	Surface water	0.00	0.00	0.00	173.08	2,411,721.00 *gal*
Application event totals		34.55	10.02	39.05	389.38	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	33.17	9.62	37.49	207.65	432,000.00 *gal*
T.I.D Canal	Surface water	0.00	0.00	0.00	134.39	1,872,633.00 *gal*
Application event totals		33.17	9.62	37.49	342.04	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	18.43	5.34	20.83	115.36	240,000.00 *gal*
T.I.D Canal	Surface water	0.00	0.00	0.00	121.03	1,686,413.00 *gal*
Application event totals		18.43	5.34	20.83	236.39	

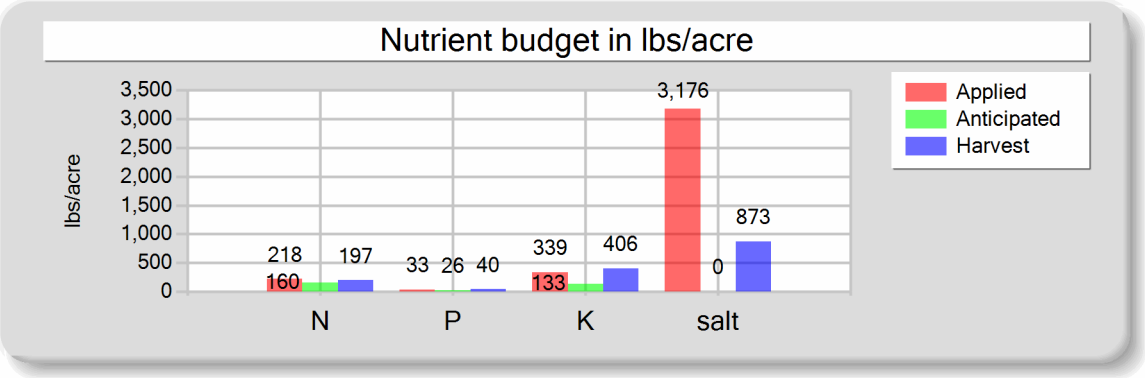
Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
T.I.D Canal	Surface water	0.00	0.00	0.00	108.29	1,508,921.00 *gal*
Application event totals		0.00	0.00	0.00	108.29	

B. NUTRIENT BUDGET

CSI - 10/18/2022: Oats, silage-soft dough

Field name: CSI | Crop: Oats, silage-soft dough | Plant date: 10/18/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	211.41	32.99	339.40	2,402.23
Fresh water	0.00	0.00	0.00	774.12
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	218.41	32.99	339.40	3,176.35
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	196.89	40.18	405.84	872.59
Nutrient balance	21.52	-7.19	-66.44	2,303.76
Applied to removed ratio	1.11	0.82	0.84	3.64

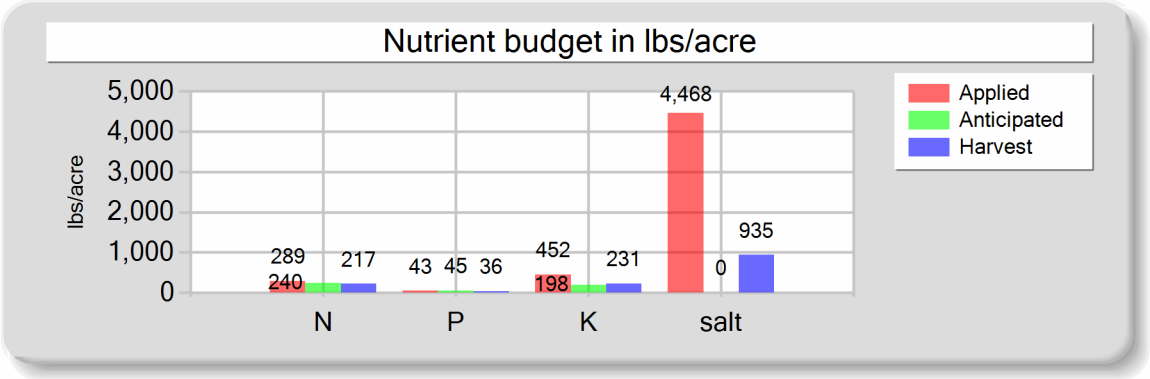
Fresh water applied	
18,984,328.00	*gallons*
699.13	*acre-inches*
15.89	*inches/acre*

Process wastewater applied	
2,964,000.00	*gallons*
109.15	*acre-inches*
2.48	*inches/acre*

Total harvests for the crop	
1	*harvests*

CSI - 06/12/2023: Corn, silage

Field name: CSI | Crop: Corn, silage | Plant date: 06/12/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	282.41	42.70	451.51	2,591.09
Fresh water	0.00	0.00	0.00	1,877.31
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	289.41	42.70	451.51	4,468.39
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	217.40	36.23	230.98	935.26
Nutrient balance	72.01	6.46	220.53	3,533.14
Applied to removed ratio	1.33	1.18	1.95	4.78

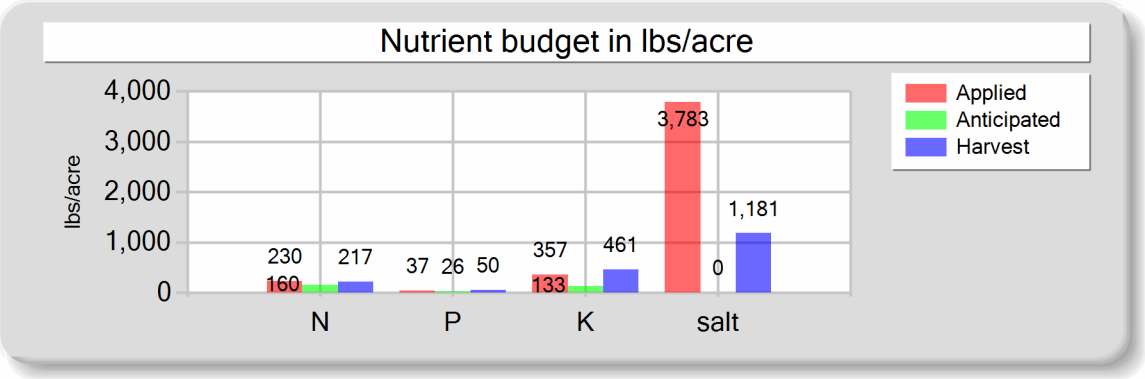
Fresh water applied	
46,038,681.00	*gallons*
1,695.45	*acre-inches*
38.53	*inches/acre*

Process wastewater applied	
3,990,000.00	*gallons*
146.94	*acre-inches*
3.34	*inches/acre*

Total harvests for the crop	
1	*harvests*

Lopes - 10/17/2022: Oats, silage-soft dough

Field name: Lopes | Crop: Oats, silage-soft dough | Plant date: 10/17/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	222.76	37.27	356.94	3,021.06
Fresh water	0.00	0.00	0.00	761.69
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	229.76	37.27	356.94	3,782.75
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	217.16	50.41	461.46	1,180.78
Nutrient balance	12.60	-13.15	-104.52	2,601.96
Applied to removed ratio	1.06	0.74	0.77	3.20

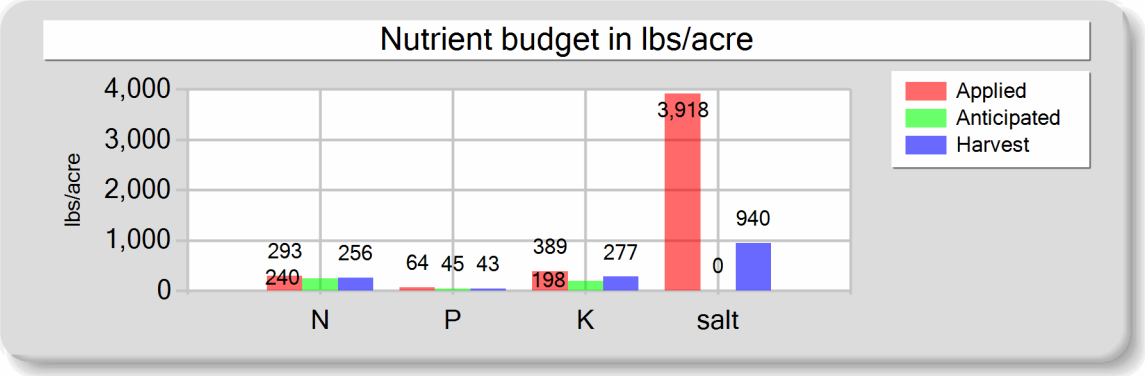
Fresh water applied	
15,283,260.00	*gallons*
562.83	*acre-inches*
15.63	*inches/acre*

Process wastewater applied	
2,862,000.00	*gallons*
105.40	*acre-inches*
2.93	*inches/acre*

Total harvests for the crop	
1	*harvests*

Lopes - 06/09/2023: Corn, silage

Field name: Lopes Crop: Corn, silage Plant date: 06/09/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	286.44	63.81	388.79	2,114.29
Fresh water	0.00	0.00	0.00	1,803.92
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	293.44	63.81	388.79	3,918.21
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	255.84	42.64	277.16	940.42
Nutrient balance	37.60	21.17	111.64	2,977.79
Applied to removed ratio	1.15	1.50	1.40	4.17

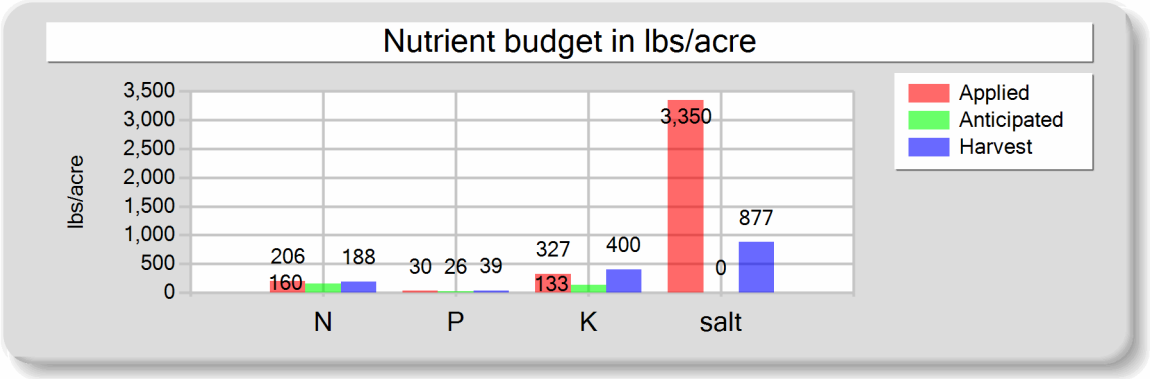
Fresh water applied	
36,195,449.00	*gallons*
1,332.96	*acre-inches*
37.03	*inches/acre*

Process wastewater applied	
4,374,000.00	*gallons*
161.08	*acre-inches*
4.47	*inches/acre*

Total harvests for the crop	
1	*harvests*

Morgan - 10/17/2022: Oats, silage-soft dough

Field name: Morgan | Crop: Oats, silage-soft dough | Plant date: 10/17/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	199.12	30.42	326.81	2,600.42
Fresh water	0.00	0.00	0.00	749.97
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	206.12	30.42	326.81	3,350.39
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	187.60	39.20	400.40	876.96
Nutrient balance	18.52	-8.78	-73.59	2,473.43
Applied to removed ratio	1.10	0.78	0.82	3.82

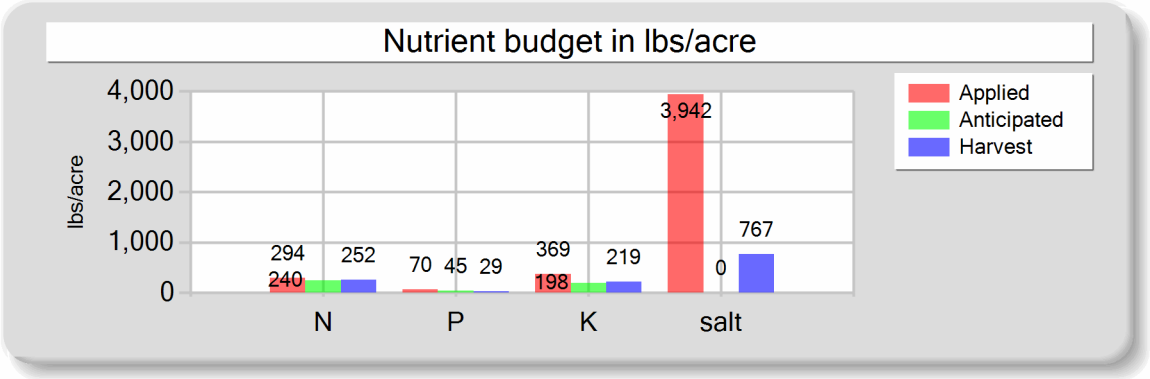
Fresh water applied	
2,926,008.00	*gallons*
107.75	*acre-inches*
15.39	*inches/acre*

Process wastewater applied	
465,000.00	*gallons*
17.12	*acre-inches*
2.45	*inches/acre*

Total harvests for the crop	
1	*harvests*

Morgan - 06/08/2023: Corn, silage

Field name: Morgan | Crop: Corn, silage | Plant date: 06/08/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	286.73	69.74	369.39	2,076.80
Fresh water	0.00	0.00	0.00	1,864.89
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	293.73	69.74	369.39	3,941.69
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	252.17	28.94	219.10	767.44
Nutrient balance	41.56	40.80	150.29	3,174.25
Applied to removed ratio	1.16	2.41	1.69	5.14

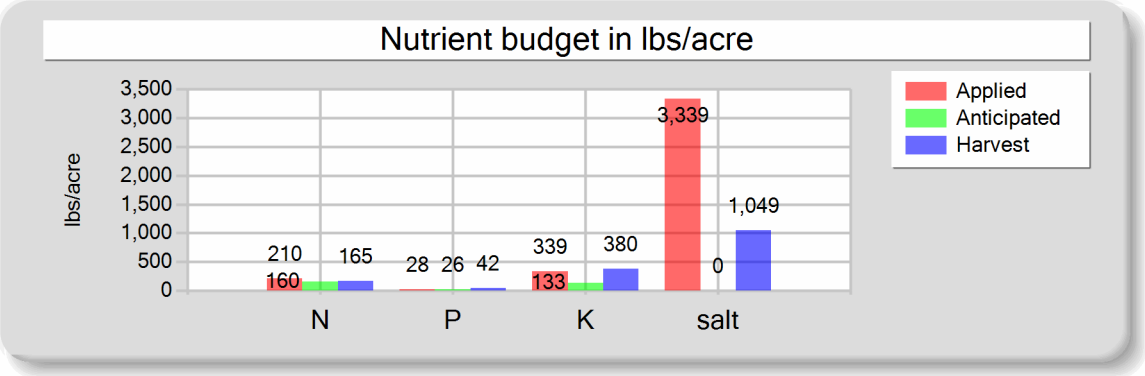
Fresh water applied	
7,275,902.00	*gallons*
267.95	*acre-inches*
38.28	*inches/acre*

Process wastewater applied	
910,500.00	*gallons*
33.53	*acre-inches*
4.79	*inches/acre*

Total harvests for the crop	
1	*harvests*

Nelson - 10/17/2022: Oats, silage-soft dough

Field name: Nelson | Crop: Oats, silage-soft dough | Plant date: 10/17/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	202.55	28.44	339.15	2,556.66
Fresh water	0.00	0.00	0.00	782.61
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	209.55	28.44	339.15	3,339.27
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	164.92	42.19	379.69	1,049.34
Nutrient balance	44.63	-13.75	-40.54	2,289.93
Applied to removed ratio	1.27	0.67	0.89	3.18

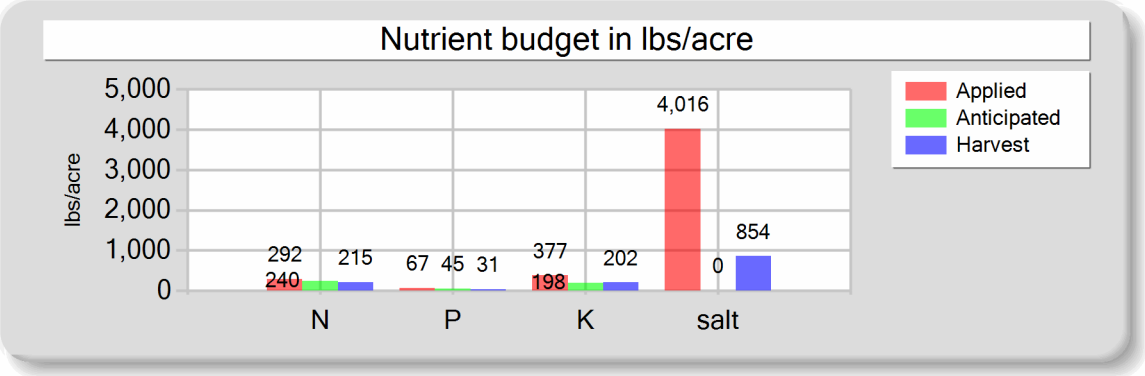
Fresh water applied	
7,415,281.00	*gallons*
273.08	*acre-inches*
16.06	*inches/acre*

Process wastewater applied	
1,080,000.00	*gallons*
39.77	*acre-inches*
2.34	*inches/acre*

Total harvests for the crop	
1	*harvests*

Nelson - 06/12/2023: Corn, silage

Field name: Nelson | Crop: Corn, silage | Plant date: 06/12/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	285.45	66.63	377.16	2,126.12
Fresh water	0.00	0.00	0.00	1,889.49
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	292.45	66.63	377.16	4,015.61
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	214.98	30.71	201.82	853.68
Nutrient balance	77.47	35.92	175.34	3,161.93
Applied to removed ratio	1.36	2.17	1.87	4.70

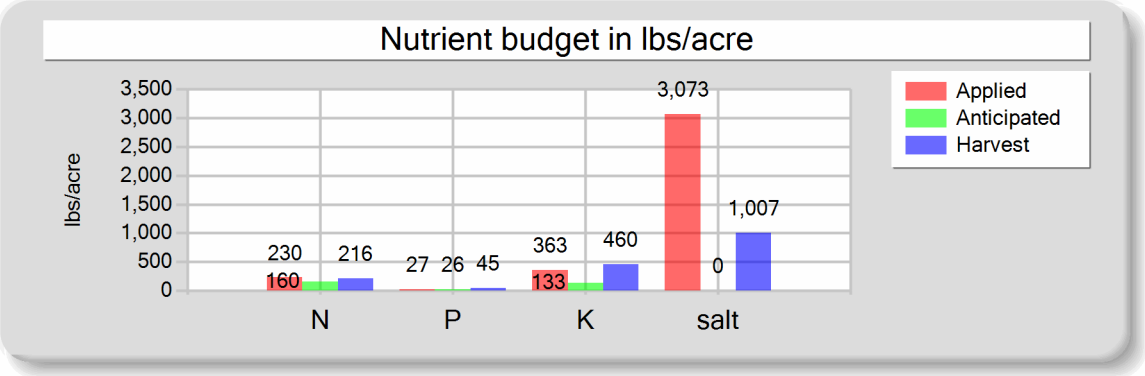
Fresh water applied	
17,903,123.00	*gallons*
659.31	*acre-inches*
38.78	*inches/acre*

Process wastewater applied	
2,133,000.00	*gallons*
78.55	*acre-inches*
4.62	*inches/acre*

Total harvests for the crop	
1	*harvests*

Perez - 10/18/2022: Oats, silage-soft dough

Field name: Perez | Crop: Oats, silage-soft dough | Plant date: 10/18/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	223.05	27.03	362.91	2,330.84
Fresh water	0.00	0.00	0.00	742.56
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	230.05	27.03	362.91	3,073.40
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	215.52	45.03	459.98	1,007.46
Nutrient balance	14.54	-18.00	-97.07	2,065.94
Applied to removed ratio	1.07	0.60	0.79	3.05

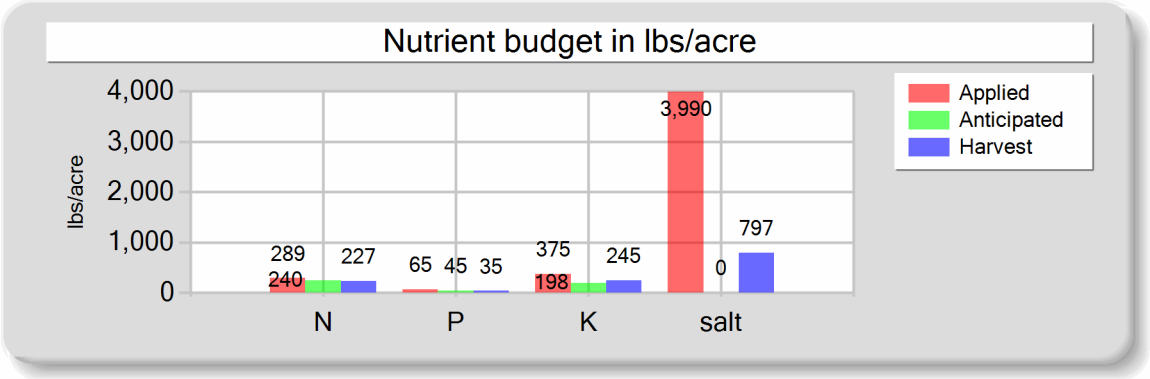
Fresh water applied	
4,966,440.00	*gallons*
182.90	*acre-inches*
15.24	*inches/acre*

Process wastewater applied	
694,500.00	*gallons*
25.58	*acre-inches*
2.13	*inches/acre*

Total harvests for the crop	
1	*harvests*

Perez - 06/08/2023: Corn, silage

Field name: Perez | Crop: Corn, silage | Plant date: 06/08/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	281.71	64.84	375.32	2,117.54
Fresh water	0.00	0.00	0.00	1,872.38
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	288.71	64.84	375.32	3,989.93
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	227.27	34.96	244.75	796.65
Nutrient balance	61.44	29.88	130.57	3,193.27
Applied to removed ratio	1.27	1.85	1.53	5.01

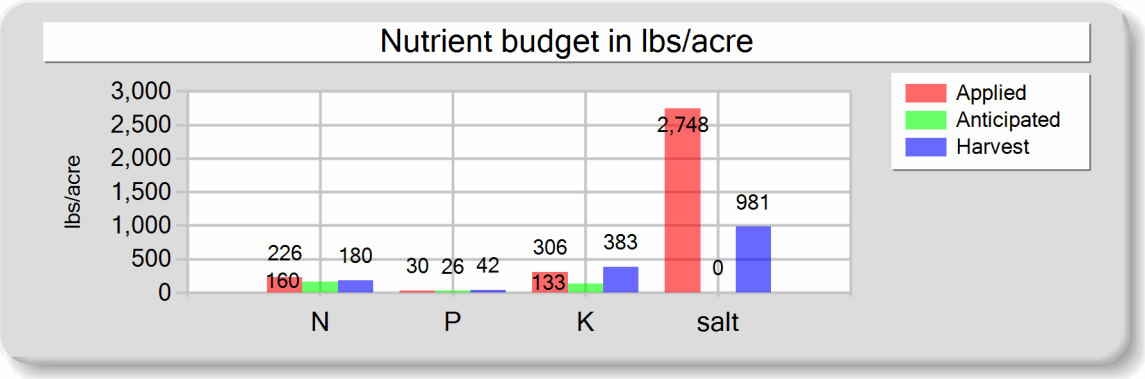
Fresh water applied	
12,523,089.00	*gallons*
461.18	*acre-inches*
38.43	*inches/acre*

Process wastewater applied	
1,470,000.00	*gallons*
54.14	*acre-inches*
4.51	*inches/acre*

Total harvests for the crop	
1	*harvests*

Rancheria - 10/17/2022: Oats, silage-soft dough

Field name: Rancheria | Crop: Oats, silage-soft dough | Plant date: 10/17/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	218.82	29.83	306.46	2,005.86
Fresh water	0.00	0.00	0.00	741.91
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	225.82	29.83	306.46	2,747.77
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	180.44	41.89	383.44	981.17
Nutrient balance	45.38	-12.06	-76.99	1,766.60
Applied to removed ratio	1.25	0.71	0.80	2.80

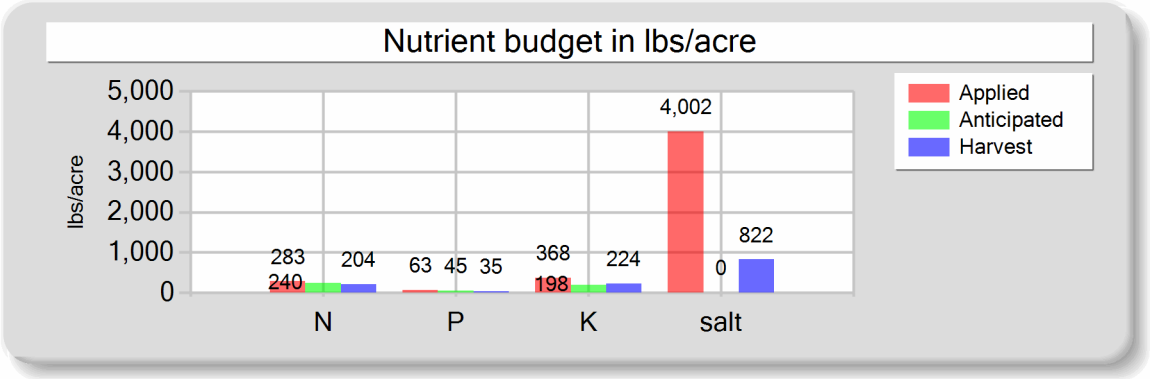
Fresh water applied	
3,721,581.00	*gallons*
137.05	*acre-inches*
15.23	*inches/acre*

Process wastewater applied	
495,000.00	*gallons*
18.23	*acre-inches*
2.03	*inches/acre*

Total harvests for the crop	
1	*harvests*

Rancheria - 06/09/2023: Corn, silage

Field name: Rancheria Crop: Corn, silage Plant date: 06/09/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	275.63	63.08	368.45	2,079.49
Fresh water	0.00	0.00	0.00	1,922.62
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	282.63	63.08	368.45	4,002.11
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	203.70	34.78	223.57	821.94
Nutrient balance	78.93	28.30	144.88	3,180.17
Applied to removed ratio	1.39	1.81	1.65	4.87

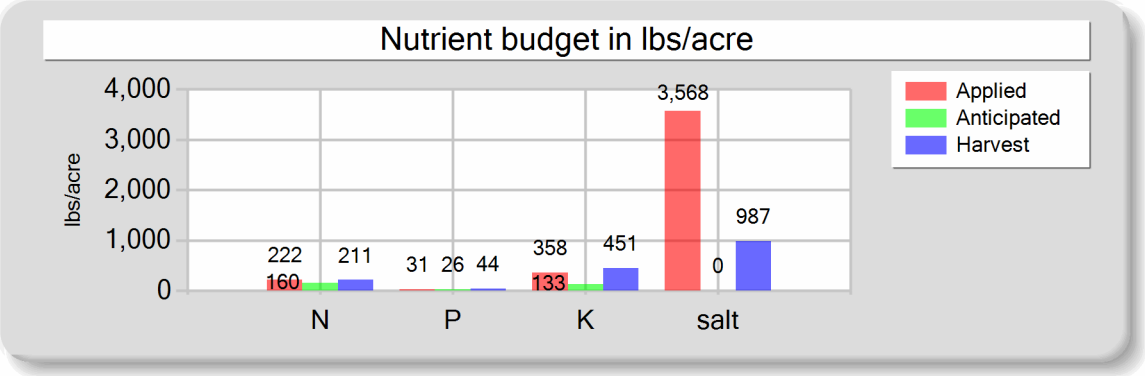
Fresh water applied	
9,644,318.00	*gallons*
355.17	*acre-inches*
39.46	*inches/acre*

Process wastewater applied	
1,074,000.00	*gallons*
39.55	*acre-inches*
4.39	*inches/acre*

Total harvests for the crop	
1	*harvests*

Randy - 10/18/2022: Oats, silage-soft dough

Field name: Randy | Crop: Oats, silage-soft dough | Plant date: 10/18/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	214.95	31.23	358.24	2,819.12
Fresh water	0.00	0.00	0.00	748.79
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	221.95	31.23	358.24	3,567.91
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	211.18	44.13	450.74	987.21
Nutrient balance	10.76	-12.90	-92.50	2,580.70
Applied to removed ratio	1.05	0.71	0.79	3.61

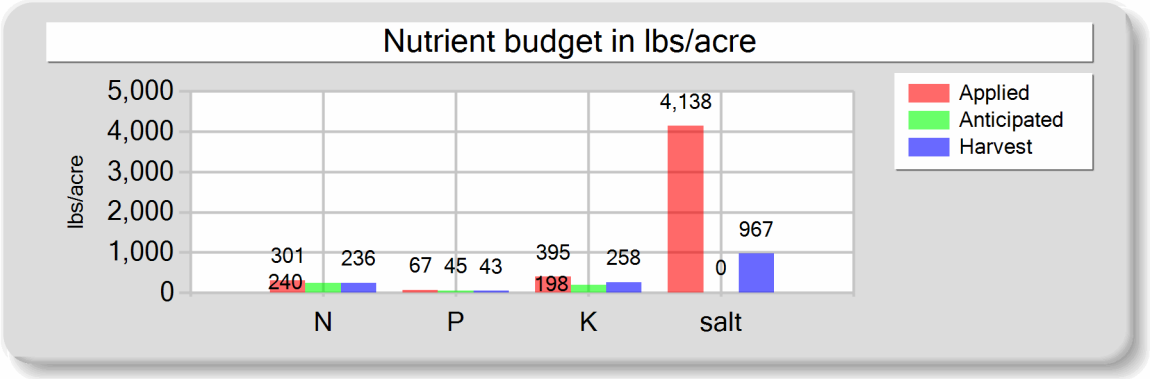
Fresh water applied	
10,433,600.00	*gallons*
384.23	*acre-inches*
15.37	*inches/acre*

Process wastewater applied	
1,749,000.00	*gallons*
64.41	*acre-inches*
2.58	*inches/acre*

Total harvests for the crop	
1	*harvests*

Randy - 06/08/2023: Corn, silage

Field name: Randy | Crop: Corn, silage | Plant date: 06/08/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	293.87	66.73	394.59	2,228.07
Fresh water	0.00	0.00	0.00	1,909.96
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	300.87	66.73	394.59	4,138.03
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	236.15	42.94	257.62	966.85
Nutrient balance	64.72	23.79	136.97	3,171.18
Applied to removed ratio	1.27	1.55	1.53	4.28

Fresh water applied	
26,613,303.00	*gallons*
980.08	*acre-inches*
39.20	*inches/acre*

Process wastewater applied	
3,162,000.00	*gallons*
116.45	*acre-inches*
4.66	*inches/acre*

Total harvests for the crop	
1	*harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

2022 Fall Separated Solids

Sample and source description: 2022 Fall Separated Solids

Sample date: 11/14/2022 Material type: Separator solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 37.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,600.00	1,500.00	1,400.00	9,700.00	1,900.00	600.00	1,600.00	563.00		82.10
DL	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00		0.01

Spring Separated Solids

Sample and source description: Spring Separated Solids

Sample date: 05/01/2023 Material type: Separator solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 55.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,400.00	2,200.00	4,500.00							75.10
DL	100.00	100.00	100.00							0.01

Fall Separated Solids

Sample and source description: Fall Separated Solids

Sample date: 10/24/2023 Material type: Separator solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 70.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	7,600.00	1,800.00	2,700.00	5,600.00	1,800.00	900.00	1,600.00	377.00		36.20
DL	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00		0.01

B. PROCESS WASTEWATER ANALYSES

2022 WWSQ4

Sample and source description: 2022 WWSQ4

Sample date: 12/12/2022 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	263.00	171.00	0.00	0.00	52.10	417.00								4,040.00	4,900
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

WWSQ1

Sample and source description: WWSQ1

Sample date: 03/13/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	297.00	176.00	0.00	0.00	65.50	418.00								4,360.00	3,640
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

WWSQ2

Sample and source description: WWSQ2

Sample date: 06/06/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	549.00	381.00	0.00	0.00	43.80	1,010.00								7,770.00	5,860
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

WWSQ3

Sample and source description: WWSQ3

Sample date: 09/19/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	230.00	123.00	0.00	0.00	66.70	260.00	78.20	56.60	63.50	1,100.00	0.00	86.10	97.10	2,610.00	1,440
DL	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	1.00	10

WWSQ4

Sample and source description: WWSQ4

Sample date: 12/11/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	322.00	302.00	0.00	0.00	35.00	491.00								5,170.00	4,520
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

C. FRESH WATER ANALYSES

T.I.D Canal

Lateral 6

Sample description: Lateral 6

Sample date: 08/28/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.00	0.00	0.00								0.00	215
DL	0.50	0.50	0.50								1.00	10

D. SOIL ANALYSES

No soil analyses entered.

E. PLANT TISSUE ANALYSES

CSI - 10/18/2022: Oats, silage-soft dough

CSI - 10/18/2022: Oats, silage-soft dough

Winter Forage

Sample and source description: Winter Forage

Sample date: 05/24/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 75.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,900.00	1,000.00	10,100.00		8.90
DL	100.00	100.00	100.00		0.01

CSI - 06/12/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/09/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 58.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,800.00	800.00	5,100.00		5.00
DL	100.00	100.00	100.00		0.01

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/09/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 62.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,000.00	600.00	3,900.00		5.80
DL	100.00	100.00	100.00		0.01

Lopes - 10/17/2022: Oats, silage-soft dough

Lopes - 10/17/2022: Oats, silage-soft dough

Winter Forage

Sample and source description: Winter Forage

Sample date: 05/24/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 65.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,600.00	1,300.00	11,900.00		8.70
DL	100.00	100.00	100.00		0.01

Lopes - 06/09/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/09/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 59.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,800.00	800.00	5,200.00		4.40
DL	100.00	100.00	100.00		0.01

Morgan - 10/17/2022: Oats, silage-soft dough

Winter Forage

Sample and source description: Winter Forage

Sample date: 05/24/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 45.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	9,400.00	2,600.00	20,200.00		8.80
DL	100.00	100.00	100.00		0.01

Morgan - 06/08/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/09/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 55.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	6,100.00	700.00	5,300.00		4.20
DL	100.00	100.00	100.00		0.01

Nelson - 06/12/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/09/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 57.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,900.00	700.00	4,600.00		4.60
DL	100.00	100.00	100.00		0.01

Perez - 10/18/2022: Oats, silage-soft dough

Winter Forage

Sample and source description: Winter Forage

Sample date: 05/24/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 65.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	6,700.00	1,400.00	14,300.00		9.00
DL	100.00	100.00	100.00		0.01

Perez - 06/08/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/09/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 62.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,200.00	800.00	5,600.00		4.90
DL	100.00	100.00	100.00		0.01

Rancheria - 10/17/2022: Oats, silage-soft dough

Winter Forage

Sample and source description: Winter Forage

Sample date: 05/24/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 65.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,300.00	1,100.00	9,900.00		8.00
DL	100.00	100.00	100.00		0.01

Rancheria - 06/09/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/09/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 62.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,100.00	700.00	4,500.00		4.40
DL	100.00	100.00	100.00		0.01

Randy - 10/18/2022: Oats, silage-soft dough

Winter Forage

Sample and source description: Winter Forage

Sample date: 05/24/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 51.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	10,900.00	2,300.00	20,100.00		8.80
DL	100.00	100.00	100.00		0.01

Randy - 06/08/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/09/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 58.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,500.00	1,000.00	6,000.00		5.40
DL	100.00	100.00	100.00		0.01

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

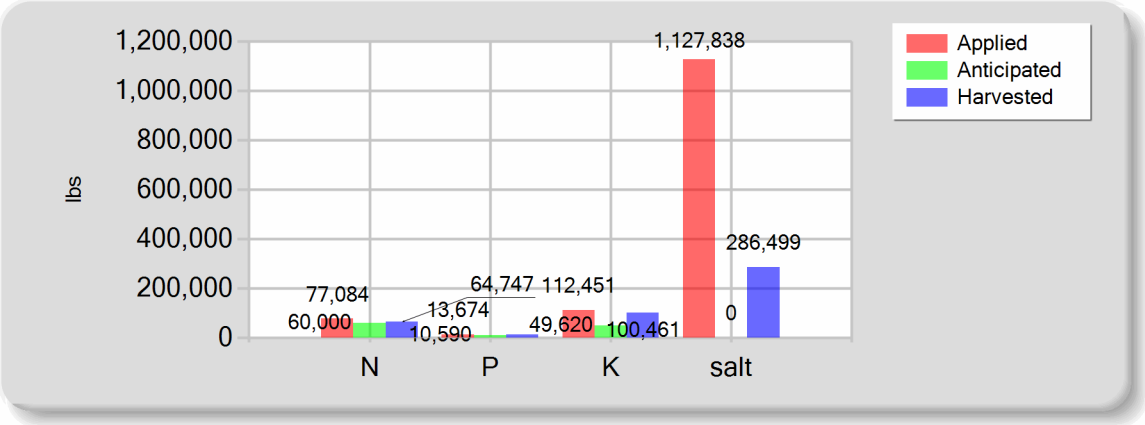
No subsurface (tile) drainage analyses entered.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

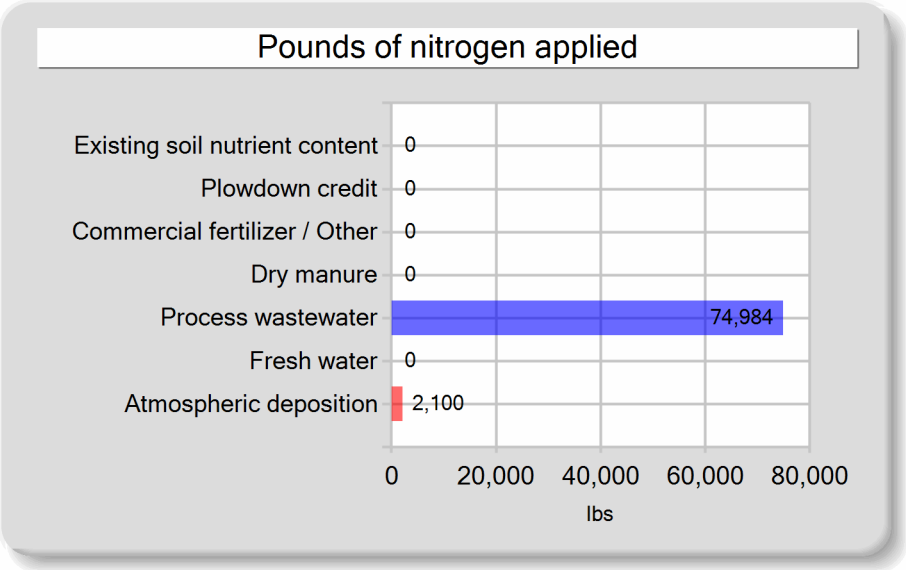
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

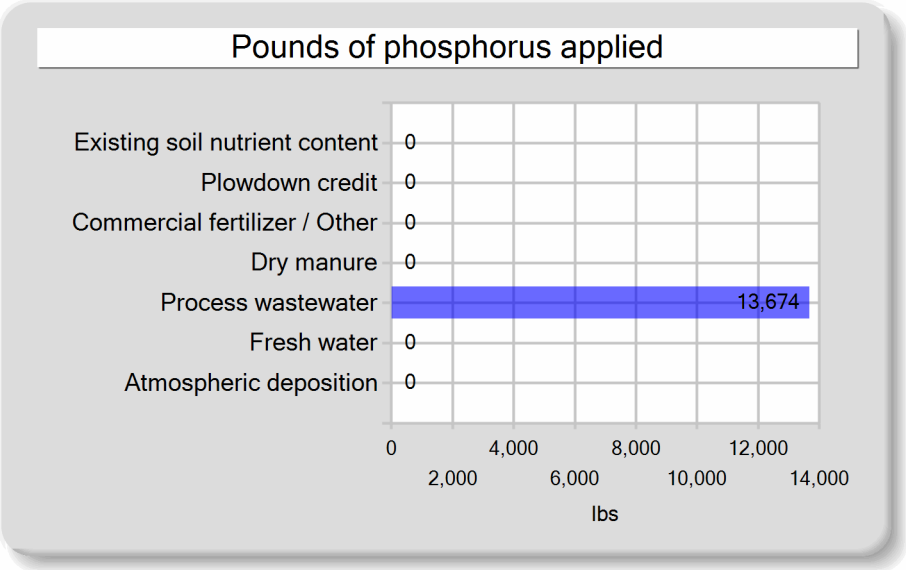
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	74,983.66	13,673.82	112,450.69	733,255.12
Fresh water	0.00	0.00	0.00	394,582.79
Atmospheric deposition	2,100.00	0.00	0.00	0.00
Total nutrients applied	77,083.66	13,673.82	112,450.69	1,127,837.92
Anticipated crop nutrient removal	60,000.00	10,590.00	49,620.00	0.00
Actual crop nutrient removal	64,747.09	12,254.90	100,461.14	286,499.20
Nutrient balance	12,336.56	1,418.92	11,989.55	841,338.72
Applied to removed ratio	1.19	1.12	1.12	3.94

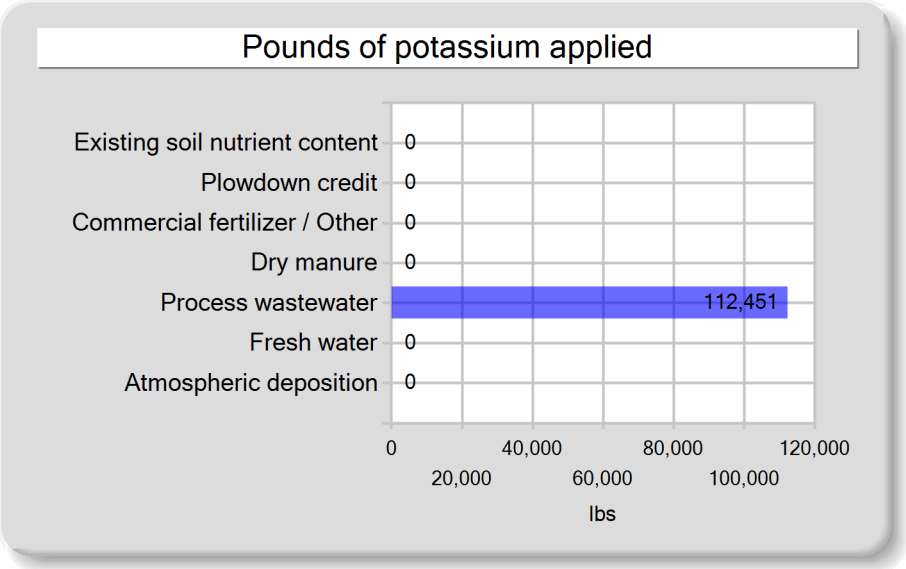
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

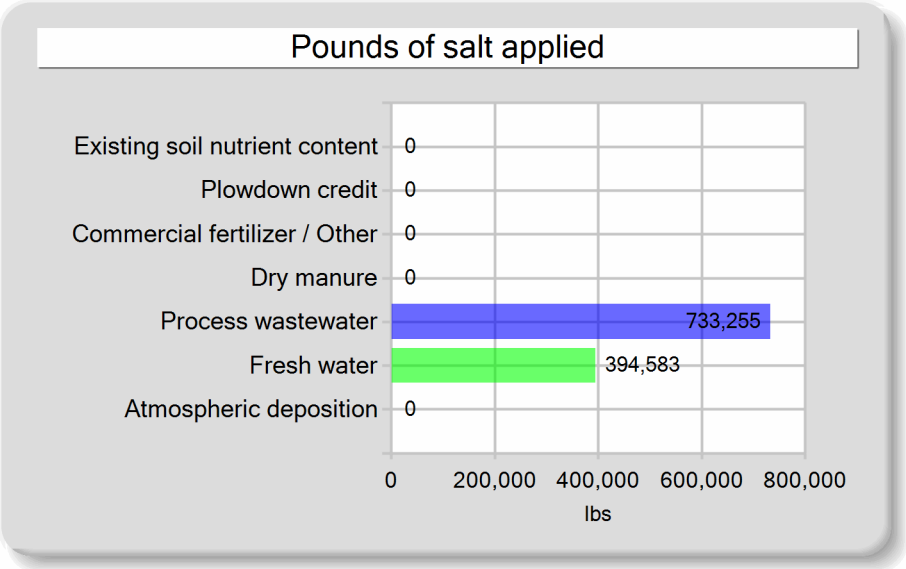


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	No
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

No notes entered for this annual report.

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

Mike Peterson Mike Peterson (May 27, 2024 15:52 PDT)	Mark Ahlem Mark Ahlem (May 28, 2024 11:27 PDT)
SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
Mike or Randy Peterson	Mark Ahlem
PRINT OR TYPE NAME	PRINT OR TYPE NAME
May 27, 2024	May 28, 2024
DATE	DATE

ATTACHMENTS

A. REQUIRED ATTACHMENTS

The following lists the required documents that should be attached to the Annual Report when submitted .

Annual Dairy Facility Assessment
Provide an Annual Dairy Facility Assessment (an update to the Preliminary Dairy Facility Assessment in Attachment A) for each reporting period. On the PDFA Final page, click on the ADFA Report button to generate an ADFA report after updating information as needed .

Manure/Process Wastewater Tracking Manifests
Provide copies of all manure/process wastewater tracking manifests for the reporting period, signed by both the owner/operator and the hauler.

Corrective Actions Documents
Provide records documenting any corrective actions taken to correct deficiencies noted as a result of the inspections required in the Monitoring Requirements of the General Order. Deficiencies not corrected in 30 days must be accompanied by an explanation of the factors preventing immediate correction.

Groundwater Monitoring
Dischargers that monitor supply wells or subsurface (tile) drainage systems, or that have monitoring well systems must submit monitoring results as directed in the General Order, Groundwater Reporting Section starting on page MRP-13.

Storm Water Monitoring
Dischargers that are required to monitor storm water more frequently than required in the General Order must submit monitoring results as directed in the General Order, Storm Water Reporting Section on page MRP-14.

ATTACHMENT D

Manure/Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

Instructions:

1) Complete one manifest for each hauling event, for each destination. A hauling event may last for several days, as long as the manure is being hauled to the same destination.
2) If there are multiple destinations, **complete a separate form for each destination**.
3) The operator must obtain the signature of the hauler upon completion of each manure-hauling event.
4) The operator shall submit copies of manure/process wastewater tracking manifest(s) with the Annual Monitoring Report for Existing Milk Cow Dairies.

Operator Information:

Name of Operator: Mark Ahlem

Name of Dairy Facility: Double Cross #6

Facility Address: 21189 American Ave. (Number and Street) Hilmar (City) 95324 (Zip Code)

Contact Person Name and Phone Number: Mark Ahlem (Name) 209-668-0867 (Phone Number)

Manure/Process Wastewater Hauler Information:

Name of Hauling Company/Person: _____

Address of Hauling Company /Person: _____ (Number and Street) _____ (City) _____ (Zip Code)

Contact Person: _____ (Name) _____ (Phone Number)

Destination Information:

Composting Facility / Broker / Farmer / Other (identify) _____ (please circle one)

Contact information of Composting Facility, Broker, Farmer, or Other (as identified above):

Double Cross #7 (Name) 20775 W. American Ave (Number and Street) Hilmar (City) 95324 (Zip Code) 209-668-0867 (Phone Number)

Manure/Process Wastewater Destination Address or Assessor's Parcel Number:

045-170-030 (Number and Street) _____ (City) _____ (Zip Code) _____ (Assessor's Parcel Number)

Dates Hauled: 3/2/23

Amount Hauled:

Enter the amount of manure hauled in tons or cubic yards (indicate the units used), the manure solids content (if amount reported in tons) or manure density (if amount reported in cubic yards), and the method used to calculate the amount:

Manure: _____ Tons or Cubic Yards (indicate which units used)

Manure Solids Content (if amount reported in tons): _____

Manure Density (if amount reported in cubic yards): _____

Method used to determine amount of manure: _____

_____

Enter the amount of process wastewater hauled in gallons and the method used to determine the amount.

Process Wastewater: 255,000 Gallons

Method used to determine volume of process wastewater: PO

_____

Written Agreement:
Does the Operator have a written agreement (in compliance with Land Application Specification C.2 of Waste Discharge Requirements General Order No. R5-2007-0035) with any party that receives process wastewater from the Operator for its own use? (please check one)

X Yes _____ No

If the answer is no, the Operator agrees to have such a written agreement with any such party for any process wastewater transferred after **31 December 2007** to such party.
_____ (Operator shall provide initials here to acknowledge this requirement).

Certification:
I declare under the penalty of law that I personally examined and am familiar with the information submitted in this document, and that based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of a fine and imprisonment for knowing violations.

Operator's Signature: Mark Ahlem (Mark Ahlem (May 28, 2024 11:27 PDT)) Date: May 28, 2024

Hauler's Signature: N/A Date: _____

ATTACHMENT D

Manure/Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

Instructions:

1) Complete one manifest for each hauling event, for each destination. A hauling event may last for several days, as long as the manure is being hauled to the same destination.
2) If there are multiple destinations, **complete a separate form for each destination**.
3) The operator must obtain the signature of the hauler upon completion of each manure-hauling event.
4) The operator shall submit copies of manure/process wastewater tracking manifest(s) with the Annual Monitoring Report for Existing Milk Cow Dairies.

Operator Information:

Name of Operator: Mark Ahlem

Name of Dairy Facility: Double Cross #6

Facility Address: 21189 American Ave. (Number and Street) Hilmar (City) 95324 (Zip Code)

Contact Person Name and Phone Number: Mark Ahlem (Name) 209-668-0867 (Phone Number)

Manure/Process Wastewater Hauler Information:

Name of Hauling Company/Person: _____

Address of Hauling Company /Person: _____ (Number and Street) _____ (City) _____ (Zip Code)

Contact Person: _____ (Name) _____ (Phone Number)

Destination Information:

Composting Facility / Broker / Farmer / Other (identify) _____ (please circle one)

Contact information of Composting Facility, Broker, Farmer, or Other (as identified above):

Double Cross #7 (Name) 20775 W. American Ave (Number and Street) Hilmar (City) 95324 (Zip Code) 209-668-0867 (Phone Number)

Manure/Process Wastewater Destination Address or Assessor's Parcel Number:

045-170-030 (Number and Street) _____ (City) _____ (Zip Code) _____ (Assessor's Parcel Number)

Dates Hauled: 4/15-4/16

Amount Hauled:

Enter the amount of manure hauled in tons or cubic yards (indicate the units used), the manure solids content (if amount reported in tons) or manure density (if amount reported in cubic yards), and the method used to calculate the amount:

Manure: _____ Tons or Cubic Yards (indicate which units used)

Manure Solids Content (if amount reported in tons): _____

Manure Density (if amount reported in cubic yards): _____

Method used to determine amount of manure: _____

_____

Enter the amount of process wastewater hauled in gallons and the method used to determine the amount.

Process Wastewater: 240,000 Gallons

Method used to determine volume of process wastewater: PO

_____

Written Agreement:
Does the Operator have a written agreement (in compliance with Land Application Specification C.2 of Waste Discharge Requirements General Order No. R5-2007-0035) with any party that receives process wastewater from the Operator for its own use? (please check one)

X Yes _____ No

If the answer is no, the Operator agrees to have such a written agreement with any such party for any process wastewater transferred after **31 December 2007** to such party.
_____ (Operator shall provide initials here to acknowledge this requirement).

Certification:
I declare under the penalty of law that I personally examined and am familiar with the information submitted in this document, and that based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of a fine and imprisonment for knowing violations.

Operator's Signature: Mark Ahlem (Mark Ahlem (May 28, 2024 11:27 PDT)) Date: May 28, 2024

Hauler's Signature: N/A Date: _____

ATTACHMENT D

Manure/Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

Instructions:

1) Complete one manifest for each hauling event, for each destination. A hauling event may last for several days, as long as the manure is being hauled to the same destination.
2) If there are multiple destinations, **complete a separate form for each destination**.
3) The operator must obtain the signature of the hauler upon completion of each manure-hauling event.
4) The operator shall submit copies of manure/process wastewater tracking manifest(s) with the Annual Monitoring Report for Existing Milk Cow Dairies.

Operator Information:

Name of Operator: Mark Ahlem

Name of Dairy Facility: Double Cross #6

Facility Address: 21189 American Ave. (Number and Street) Hilmar (City) 95324 (Zip Code)

Contact Person Name and Phone Number: Mark Ahlem (Name) 209-668-0867 (Phone Number)

Manure/Process Wastewater Hauler Information:

Name of Hauling Company/Person: _____

Address of Hauling Company /Person: _____ (Number and Street) _____ (City) _____ (Zip Code)

Contact Person: _____ (Name) _____ (Phone Number)

Destination Information:

Composting Facility / Broker / Farmer / Other (identify) _____ (please circle one)

Contact information of Composting Facility, Broker, Farmer, or Other (as identified above):

Double Cross #7 (Name) 20775 W. American Ave (Number and Street) Hilmar (City) 95324 (Zip Code) 209-668-0867 (Phone Number)

Manure/Process Wastewater Destination Address or Assessor's Parcel Number:

_____ (Number and Street) _____ (City) _____ (Zip Code) 045-170-030 (Assessor's Parcel Number)

Dates Hauled: 7/22-7/23

Amount Hauled:

Enter the amount of manure hauled in tons or cubic yards (indicate the units used), the manure solids content (if amount reported in tons) or manure density (if amount reported in cubic yards), and the method used to calculate the amount:

Manure: _____ Tons or Cubic Yards (indicate which units used)

Manure Solids Content (if amount reported in tons): _____

Manure Density (if amount reported in cubic yards): _____

Method used to determine amount of manure: _____

_____

Enter the amount of process wastewater hauled in gallons and the method used to determine the amount.

Process Wastewater: 510,000 Gallons

Method used to determine volume of process wastewater: PO

_____

Written Agreement:
Does the Operator have a written agreement (in compliance with Land Application Specification C.2 of Waste Discharge Requirements General Order No. R5-2007-0035) with any party that receives process wastewater from the Operator for its own use? (please check one)

X Yes _____ No

If the answer is no, the Operator agrees to have such a written agreement with any such party for any process wastewater transferred after **31 December 2007** to such party.
_____ (Operator shall provide initials here to acknowledge this requirement).

Certification:
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Operator's Signature: Mark Ahlem (Mark Ahlem (May 28, 2024 11:27 PDT)) Date: May 28, 2024

Hauler's Signature: NA Date: _____

ATTACHMENT D

Manure/Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

Instructions:

1) Complete one manifest for each hauling event, for each destination. A hauling event may last for several days, as long as the manure is being hauled to the same destination.
2) If there are multiple destinations, **complete a separate form for each destination**.
3) The operator must obtain the signature of the hauler upon completion of each manure-hauling event.
4) The operator shall submit copies of manure/process wastewater tracking manifest(s) with the Annual Monitoring Report for Existing Milk Cow Dairies.

Operator Information:

Name of Operator: Mark Ahlem

Name of Dairy Facility: Double Cross #6

Facility Address: 21189 American Ave. (Number and Street) Hilmar (City) 95324 (Zip Code)

Contact Person Name and Phone Number: Mark Ahlem (Name) 209-668-0867 (Phone Number)

Manure/Process Wastewater Hauler Information:

Name of Hauling Company/Person: _____

Address of Hauling Company /Person: _____ (Number and Street) _____ (City) _____ (Zip Code)

Contact Person: _____ (Name) _____ (Phone Number)

Destination Information:

Composting Facility / Broker / Farmer / Other (identify) _____ (please circle one)

Contact information of Composting Facility, Broker, Farmer, or Other (as identified above):

Double Cross #7 (Name) 20775 W. American Ave (Number and Street) Hilmar (City) 95324 (Zip Code) 209-668-0867 (Phone Number)

Manure/Process Wastewater Destination Address or Assessor's Parcel Number:

_____ (Number and Street) _____ (City) _____ (Zip Code) 045-170-030 (Assessor's Parcel Number)

Dates Hauled: 8/25/23

Amount Hauled:

Enter the amount of manure hauled in tons or cubic yards (indicate the units used), the manure solids content (if amount reported in tons) or manure density (if amount reported in cubic yards), and the method used to calculate the amount:

Manure: _____ Tons or Cubic Yards (indicate which units used)

Manure Solids Content (if amount reported in tons): _____

Manure Density (if amount reported in cubic yards): _____

Method used to determine amount of manure: _____

_____

Enter the amount of process wastewater hauled in gallons and the method used to determine the amount.

Process Wastewater: 744,000 Gallons

Method used to determine volume of process wastewater: PO

_____

Written Agreement:
Does the Operator have a written agreement (in compliance with Land Application Specification C.2 of Waste Discharge Requirements General Order No. R5-2007-0035) with any party that receives process wastewater from the Operator for its own use? (please check one)

X Yes _____ No

If the answer is no, the Operator agrees to have such a written agreement with any such party for any process wastewater transferred after **31 December 2007** to such party.
_____ (Operator shall provide initials here to acknowledge this requirement).

Certification:
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Operator's Signature: Mark Ahlem (Mark Ahlem (May 28, 2024 11:27 PDT)) Date: May 28, 2024

Hauler's Signature: N/A Date: _____

ATTACHMENT D

Manure/Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

Instructions:

1) Complete one manifest for each hauling event, for each destination. A hauling event may last for several days, as long as the manure is being hauled to the same destination.
2) If there are multiple destinations, **complete a separate form for each destination.**
3) The operator must obtain the signature of the hauler upon completion of each manure-hauling event.
4) The operator shall submit copies of manure/process wastewater tracking manifest(s) with the Annual Monitoring Report for Existing Milk Cow Dairies.

Operator Information:

Name of Operator: Mark Ahlem

Name of Dairy Facility: Double Cross #6

Facility Address: 21189 American Ave. (Number and Street) Hilmar (City) 95324 (Zip Code)

Contact Person Name and Phone Number: Mark Ahlem (Name) 209-668-0867 (Phone Number)

Manure/Process Wastewater Hauler Information:

Name of Hauling Company/Person: _____

Address of Hauling Company /Person: _____ (Number and Street) _____ (City) _____ (Zip Code)

Contact Person: _____ (Name) _____ (Phone Number)

Destination Information:

Composting Facility / Broker / Farmer / Other (identify) _____ (please circle one)

Contact information of Composting Facility, Broker, Farmer, or Other (as identified above):

Double Cross #7 (Name) 20775 W. American Ave (Number and Street) Hilmar (City) 95324 (Zip Code) 209-668-0867 (Phone Number)

Manure/Process Wastewater Destination Address or Assessor's Parcel Number:

_____ (Number and Street) _____ (City) _____ (Zip Code) 045-170-030 (Assessor's Parcel Number)

Dates Hauled: 9/3/23

Amount Hauled:

Enter the amount of manure hauled in tons or cubic yards (indicate the units used), the manure solids content (if amount reported in tons) or manure density (if amount reported in cubic yards), and the method used to calculate the amount:

Manure: _____ Tons or Cubic Yards (indicate which units used)

Manure Solids Content (if amount reported in tons): _____

Manure Density (if amount reported in cubic yards): _____

Method used to determine amount of manure: _____

_____

Enter the amount of process wastewater hauled in gallons and the method used to determine the amount.

Process Wastewater: 390,000 Gallons

Method used to determine volume of process wastewater: PO

_____

Written Agreement:
Does the Operator have a written agreement (in compliance with Land Application Specification C.2 of Waste Discharge Requirements General Order No. R5-2007-0035) with any party that receives process wastewater from the Operator for its own use? (please check one)

X Yes _____ No

If the answer is no, the Operator agrees to have such a written agreement with any such party for any process wastewater transferred after **31 December 2007** to such party.
_____ (Operator shall provide initials here to acknowledge this requirement).

Certification:
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Operator's Signature: Mark Ahlem (Mark Ahlem (May 28, 2024 11:27 PDT)) Date: May 28, 2024

Hauler's Signature: NA Date: _____

ATTACHMENT D

Manure/Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

Instructions:

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2) If there are multiple destinations, **complete a separate form for each destination**.
3) The operator must obtain the signature of the hauler upon completion of each manure-hauling event.
4) The operator shall submit copies of manure/process wastewater tracking manifest(s) with the Annual Monitoring Report for Existing Milk Cow Dairies.

Operator Information:

Name of Operator: Mark Ahlem

Name of Dairy Facility: Double Cross #6

Facility Address: 21189 American Ave. (Number and Street) Hilmar (City) 95324 (Zip Code)

Contact Person Name and Phone Number: Mark Ahlem (Name) 209-668-0867 (Phone Number)

Manure/Process Wastewater Hauler Information:

Name of Hauling Company/Person: _____

Address of Hauling Company /Person: _____ (Number and Street) _____ (City) _____ (Zip Code)

Contact Person: _____ (Name) _____ (Phone Number)

Destination Information:

Composting Facility / Broker / Farmer / Other (identify) _____ (please circle one)

Contact information of Composting Facility, Broker, Farmer, or Other (as identified above):

Double Cross #7 (Name) 20775 W. American Ave (Number and Street) Hilmar (City) 95324 (Zip Code) 209-668-0867 (Phone Number)

Manure/Process Wastewater Destination Address or Assessor's Parcel Number:

_____ (Number and Street) _____ (City) _____ (Zip Code) 045-170-030 (Assessor's Parcel Number)

Dates Hauled: 9/18/23

Amount Hauled:

Enter the amount of manure hauled in tons or cubic yards (indicate the units used), the manure solids content (if amount reported in tons) or manure density (if amount reported in cubic yards), and the method used to calculate the amount:

Manure: _____ Tons or Cubic Yards (indicate which units used)

Manure Solids Content (if amount reported in tons): _____

Manure Density (if amount reported in cubic yards): _____

Method used to determine amount of manure: _____

_____

Enter the amount of process wastewater hauled in gallons and the method used to determine the amount.

Process Wastewater: 390,000 Gallons

Method used to determine volume of process wastewater: PO

_____

Written Agreement:
Does the Operator have a written agreement (in compliance with Land Application Specification C.2 of Waste Discharge Requirements General Order No. R5-2007-0035) with any party that receives process wastewater from the Operator for its own use? (please check one)

X Yes _____ No

If the answer is no, the Operator agrees to have such a written agreement with any such party for any process wastewater transferred after **31 December 2007** to such party.
_____ (Operator shall provide initials here to acknowledge this requirement).

Certification:
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Operator's Signature: Mark Ahlem (Mark Ahlem (May 28, 2024 11:27 PDT)) Date: May 28, 2024

Hauler's Signature: N/A Date: _____

ATTACHMENT D

Manure/Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

Instructions:

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2) If there are multiple destinations, **complete a separate form for each destination**.
3) The operator must obtain the signature of the hauler upon completion of each manure-hauling event.
4) The operator shall submit copies of manure/process wastewater tracking manifest(s) with the Annual Monitoring Report for Existing Milk Cow Dairies.

Operator Information:

Name of Operator: Mark Ahlem

Name of Dairy Facility: Double Cross #6

Facility Address: 21189 American Ave. (Number and Street) Hilmar (City) 95324 (Zip Code)

Contact Person Name and Phone Number: Mark Ahlem (Name) 209-668-0867 (Phone Number)

Manure/Process Wastewater Hauler Information:

Name of Hauling Company/Person: Dores Ag Services, Inc

Address of Hauling Company /Person: P.O Box 65 (Number and Street) Stevinson (City) 95374 (Zip Code)

Contact Person: Phil Dores (Name) 209-649-4503 (Phone Number)

Destination Information:

Composting Facility / Broker / Farmer / Other (identify) _____ (please circle one)

Contact information of Composting Facility, Broker, Farmer, or Other (as identified above):

Charles Ahlem Ranches (Name) 23546 American Ave (Number and Street) Hilmar (City) 95324 (Zip Code) 209-668-0867 (Phone Number)

Manure/Process Wastewater Destination Address or Assessor's Parcel Number:

045-020-032 (Number and Street / City / Zip Code / Assessor's Parcel Number)

Dates Hauled: 5/27/23

Amount Hauled:

Enter the amount of manure hauled in tons or cubic yards (indicate the units used), the manure solids content (if amount reported in tons) or manure density (if amount reported in cubic yards), and the method used to calculate the amount:

Manure: 2000 Tons or Cubic Yards (indicate which units used)

Manure Solids Content (if amount reported in tons): @ 55%

Manure Density (if amount reported in cubic yards): _____

Method used to determine amount of manure: @12 ton per load

Enter the amount of process wastewater hauled in gallons and the method used to determine the amount.

Process Wastewater: _____ Gallons

Method used to determine volume of process wastewater: _____

Written Agreement:
Does the Operator have a written agreement (in compliance with Land Application Specification C.2 of Waste Discharge Requirements General Order No. R5-2007-0035) with any party that receives process wastewater from the Operator for its own use? (please check one)

____ Yes _____ No

If the answer is no, the Operator agrees to have such a written agreement with any such party for any process wastewater transferred after **31 December 2007** to such party.
_____ (Operator shall provide initials here to acknowledge this requirement).

Certification:
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Operator's Signature: Mark Ahlem (Mark Ahlem (May 28, 2024 11:27 PDT)) Date: May 28, 2024

Hauler's Signature: _____ Date: _____

Denele Analytical, Inc.

865 South Kilroy Avenue Turlock, CA 95380
Phone (209) 634-9055 - Fax (209) 634-9057

www.denelelabs.com

40 N. East Street, Woodland, CA 95776
Phone (530) 666-9056 - Fax (530) 666-9058

Compliance Analysis Report

Customer
Double Cross #1
23546 W. American Ave.
Hilmar, CA 95324

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER: #6
RECEIVED DATE: 10/18/23 9:45 AM
SUBMITTED BY: KJ
ANALYZED DATE: 10/19/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 15.0c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
DW 6	T3291017A	Electrical Conductivity (EC)	EPA 120.1 and	1.22	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	44.4	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	780	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	101	96.4 (T3291016A)	97.6 (T3291016A)	1.23
Electrical Conductivity (EC)	97.6			0.190
Nitrate Nitrogen (NO3-N)	99.2	103 (W3291003A)	99.9 (W3291003A)	2.88

The warranty of Denele Analytical is limited to the accuracy of the analyses of the samples as received. Denele Analytical assumes no responsibility for which the customer uses our test results, nor liability for any other warranties, express or implied. These terms and conditions shall supercede any conflicting terms and conditions submitted on customer purchase orders or other forms submitted for work.

ELAP Certificate No.: 2714

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Compliance Analysis Report

Customer
Double Cross #1
23546 W. American Ave.
Hilmar, CA 95324

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER: #6
RECEIVED DATE: 10/18/23 9:45 AM
SUBMITTED BY: KJ
ANALYZED DATE: 10/19/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 14.0c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
DW 9	T3291017B	Electrical Conductivity (EC)	EPA 120.1 and	0.968	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	5.08	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	619	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	101	96.4 (T3291016A)	97.6 (T3291016A)	1.23
Electrical Conductivity (EC)	97.6			0.190
Nitrate Nitrogen (NO3-N)	99.2	103 (W3291003A)	99.9 (W3291003A)	2.88

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Compliance Analysis Report

Customer
Double Cross #1
23546 W. American Ave.
Hilmar, CA 95324

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER: #6
RECEIVED DATE: 10/18/23 9:45 AM
SUBMITTED BY: KJ
ANALYZED DATE: 10/19/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 14.0c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
DW 5	T3291017C	Electrical Conductivity (EC)	EPA 120.1 and	0.744	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	4.53	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	476	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	101	96.4 (T3291016A)	97.6 (T3291016A)	1.23
Electrical Conductivity (EC)	97.6			0.190
Nitrate Nitrogen (NO3-N)	99.2	103 (W3291003A)	99.9 (W3291003A)	2.88

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Phone (530) 666-9056 - Fax (530) 666-9058

Compliance Analysis Report

Customer
Double Cross #1
23546 W. American Ave.
Hilmar, CA 95324

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER: #6
RECEIVED DATE: 10/18/23 9:45 AM
SUBMITTED BY: KJ
ANALYZED DATE: 10/19/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 14.2c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
DW 10	T3291017D	Electrical Conductivity (EC)	EPA 120.1 and	0.705	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	26.4	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	451	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	101	96.4 (T3291016A)	97.6 (T3291016A)	1.23
Electrical Conductivity (EC)	97.6			0.190
Nitrate Nitrogen (NO3-N)	99.2	103 (W3291003A)	99.9 (W3291003A)	2.88

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Compliance Analysis Report

Customer
Double Cross #1
23546 W. American Ave.
Hilmar, CA 95324

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
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RECEIVED DATE: 10/18/23 9:45 AM
SUBMITTED BY: KJ
ANALYZED DATE: 10/19/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 14.0c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
DW 7	T3291017E	Electrical Conductivity (EC)	EPA 120.1 and	0.969	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	5.08	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	620	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	101	96.4 (T3291016A)	97.6 (T3291016A)	1.23
Electrical Conductivity (EC)	97.6			0.190
Nitrate Nitrogen (NO3-N)	99.2	103 (W3291003A)	99.9 (W3291003A)	2.88

The warranty of Denele Analytical is limited to the accuracy of the analyses of the samples as received. Denele Analytical assumes no responsibility for which the customer uses our test results, nor liability for any other warranties, express or implied. These terms and conditions shall supercede any conflicting terms and conditions submitted on customer purchase orders or other forms submitted for work.

DENELE ANALYTICAL, INC.

865 S. KILROY
TURLOCK, CA 95380

PH# 209-634-9055 FAX# 209-634-9057

DAIRY COMPLIANCE

Client: DC#6
Mailing address:
(not facility address)
Phone#:

Authorized Copy Released To: FR
Sampled By: SB
Project ID:

ANALYSIS TO BE COMPLETED:

IRRIGATION/GROUND WATER
- H1 EC*, NO3-N, NH4-N*
- H6 EC, TN, TDS
- H7 Combination of H1 & H6
- H14 EC, NO3-N, NH4-N, Ca, Mg, Na, Bicarb Carb, SO4, Cl, TDS

PROCESS WASTE WATER
- H3 EC*, NH4-N, TKN, TP, K, TDS
- H4 EC*, NO3-N, NH4-N, TKN, TP, K, TDS
- H5 Ca, Mg, Na, Bicarb, Carb, SO4, Cl
- H8 Combination of H3 &H5
- H9 Combination of H4 &H5

TILE DRAIN
- H2 EC*, NO3-N, TP, NH4-N*, TDS

** Required Field Measurement*

___ (Check here if performed)

DISCHARGE WATER
- E1 EC*, Temp*, pH*, NO3-N, NH4-N, TKN, TP, K, TDS, Dissolved Oxygen*, BOD 5, TSS, Total & Fecal Coliforms
- E2 EC,* Temp*, pH*, NO3-N, NH4-N*, Turbidity, TP, Dissolved Oxygen*, Total & Fecal Coliforms
- E3 EC*, Temp*, pH*, NO3-N, NH4-N*, TP, Total & Fecal Coliforms
- E4 EC*, Temp*, pH*, NO3-N, NH4-N*, Turbidity, TP, Total & Fecal Coliform

MANURE
- M1 % Moisture
- M3 TN TP, K, % Moisture
- M4 Ca, Mg, Na, Bicarb, Carb, SO4, Cl, Ash
- M7 Combination of M3 & M4

PLANT TISSUE
- F18 % Moisture, Ash, TN, P, K, DM and as received
- F18+ % Moisture, Ash, TN, P, K, DM and as received (Package also includes NIR DCAD)
- F19+ % Moisture, TN (Package includes complete plant tissue analysis)

SOIL
- K3 Soluble Phosphorus
- K4 1ft NO3, Organic Matter
- K5 Pre-Plant (2ft NO3), Fall (1-2ft NO3), Fall (2-3ft NO3)
- K6 0-1ft EC, NO3, Soluble P, K, Organic Matter
- K7 Combination of K3 & K4
- K8 Combination of K3 & K5

	Date Collected	Time Collected	Sample ID	Description	Temp	Well Purge	H1	H3	M3	F18	OTHER
1	10/10	12:30	Domestic	DW 6	15.0'		✓				
2		12:45	well	DW 9	14.0'		✓				
3		1:00		DW 5	14.0'		✓				
4		1:15		DW 10	14.2'		✓				
5	✓	1:30	✓	DW 7	14.0'		✓				
6											
7											
8											
9											
10											

Analysis

CHAIN OF CUSTODY RECORDING

	Signature	Company	Received Date & Time	Relinquished Date & Time	Logged in By:	LABORATORY NUMBER
1st		FRAG				
2nd		DENELE ANALYTICAL, INC.	10-18-23 9:45			T3291017

(LABORATORY USE ONLY)

Required to be submitted to any agency? (YES / NO)

2023 Annual Report Double Cross #6

Final Audit Report 2024-05-28

Created:	2024-05-21
By:	Kristen Jones (kjones@fragservices.com)
Status:	Signed
Transaction ID:	CBJCHBCAABAA5PEcGSFpS4i2nYVXCCGOv46HS6JvybeQ

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